Adjusted R-Square

In regression models, especially **Linear Regression**, we use **R^2 (R-squared)** to measure how well the model explains the variability of the target variable.

R^2 (Coefficient of Determination)

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

- **SSres** = Sum of Squared Residuals (errors)
- **SStot** = Total Sum of Squares

Problem with R^2:

R^2 tells us "What percentage of the variance in the target variable is explained by the model?"

Example:
$R^2 = 0.80 \rightarrow$ Model explains 80% of the variance.

Adjusted R-Square

Problem with R^2

If you keep adding more features (even useless ones):

- R^2 **never decreases**
- It either **increases or stays the same**

Why ?

R^2 (Coefficient of Determination)

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

- **SSres** = Sum of Squared Residuals (errors)
- **SStot** = Total Sum of Squares

$$y = b_0 + b_1 x_1 + b_2 x_2 + \underbrace{b_3 x_3}_{\text{new term}}$$

$$SS_{tot} = \sum \underbrace{(y_i - \bar{y})^2}_{\text{No change}}$$

because y_i, $\bar{y}$ remain same

$$SS_{res} = \sum (y_i - \hat{y}_i)^2$$

decreases or stays same

$$\Rightarrow R^2 = 1 - \frac{SS_{res}}{SS_{tot}} = \text{increases or stays same}$$

So, by adding more features

- **R^2** either **increases or stays the same**

$$y = b_0 + b_1 x_1 + b_2 x_2 + b_3 x_3$$

This may encourage **overfitting**

This is where **Adjusted R^2** comes in.

Adjusted R^2 corrects R^2 by penalizing unnecessary predictors.

Formula:

$$\text{Adjusted } R^2 = 1 - \left(\frac{(1 - R^2)(n - 1)}{n - p - 1}\right)$$

Where:

n = number of observations

p = number of predictors (features)

R^2 = regular R-squared

$p \uparrow \Rightarrow (n-p-1) \downarrow \Rightarrow \frac{n-1}{n-p-1} \uparrow$ (for const. R) $\Rightarrow$ Adj. $R^2 \downarrow$

$\Rightarrow$ (Sig. increase in R)

Adj. $R^2 \uparrow$

Key points of **adjusted R^2:**

- Increases only if new feature improves model significantly
- Decreases if new feature adds little or no value
- Rewards good features and penalizes unnecessary features

$$y = b_0 + b_1 x_1 + b_2 x_2 + b_3 x_3$$

Simple Example

Model	Features	R^2	Adjusted R^2	
Model 1	2 features	0.80	0.78	x_1, x_2
Model 2	5 features	0.85	0.74	x_1, x_2, x_3, x_4, x_5

Even though R^2 increased (0.80 → 0.85), but **adjusted R^2** dropped (0.78 → 0.74).

This means extra features are probably useless.

Adjusted R-Square

When Should You Use Adjusted R^2?

Use **Adjusted R^2** when:

- Comparing **multiple regression models**
- Feature selection

STOP

Fhdsklf
Fjdsklf
Fjskldf